PriceWise Pipeline — What's Been Done

The full pipeline in `pricewise_pipeline.ipynb` now runs end to end on the Canada valid-sales file. It ends with a recommended price for each of 30 pilot products: 10 increases, 10 decreases and 10 holds. The frozen baseline forecasts 28-day demand with 45.3% WAPE, slightly better than the best simple benchmark (45.9%).

Pipeline at a glance

Input is `CA_valid_sales_all_transactions.csv`: 498,263 Canada enterprise Standard Invoice lines, Apr 2023 – Aug 2026. Each stage reads the previous stage's saved outputs.

Stage	Steps	What it produced
1 Data preparation	Clean & label → product eligibility → family diagnostics → pilot 30 → daily panel	118 of 3,715 products eligible; 30 pilot products; daily panel of 30 products × 1,246 days
1.5 Global Baseline v1	Simple benchmarks → learn global weights → time-based validation → freeze	`baseline_v1.json`: 45.3% WAPE vs 45.9% for the best benchmark
1.8 Baseline v2	Test 5 segmentation schemes → compare weights → profiles → freeze	`baseline_v2.json`: no scheme passed, so v2 = one global profile
2 Pricing	Business signals → objective weights → price adjustment → guardrails → final price	`s2_price_recommendations.csv`: 10 up, 10 down, 10 hold

The baseline answers "how many units will this product sell in the next 28 days at today's price?" Stage 2 then asks what happens to margin, revenue and volume if the price moves, and picks the best move within safe limits.

Stage 1 — Data preparation

Clean and label

No rows are deleted. Every line gets label columns, and later steps filter on them.

Label	Lines	Rule
Bundled price	39,826	Unit price under $1, mostly $0.01 Services lines
Negative margin	89,671	margin < 0 (kept, flagged)
Coverage gap	16,572	Line comes from the FY26Q3 file (Oct–Dec 2025)
Exact duplicate	13,909	Kept, per the data-prep note
Price outlier	7,138	Outside 5× IQR of the product's own log unit price
Zero cost	852	cost ⩽ 0
Non-priceable type	111	Shipping, Other or missing product type

Coverage gap. The FY26Q3 file (Oct–Dec 2025) has about 2 active customers a week, against about 25 in every other quarter, and units fall 60–75%. This is missing data, not lost demand. Those days are set to unknown (not zero), are left out of rolling averages, and no forecast that touches them is scored.

Revenue looks like USD: `invoice_revenue_country_currency` is about 1.37× `revenue`.

Product eligibility (118 of 3,715)

A product-market is product × country. There are two lifecycle tiers, because laptop models sell for only about 12–15 months, so a single 24-month rule removed every notebook.

Test	Run-rate (Options, Services)	Lifecycle (Notebook, Workstation, Desktop, Visuals, Servers)
Active months	⩾ 24	⩾ 9
First sale by	Aug 2024	Sep 2025
Total units	⩾ 500	⩾ 300

Test	Run-rate (Options, Services)	Lifecycle (Notebook, Workstation, Desktop, Visuals, Servers)
Sold in last 90 days	yes	yes
Bundled-price units	⩽ 10%	⩽ 10%
Distinct price points	⩾ 10	⩾ 10
Price CV	0.03–0.60	0.03–0.60

Main reasons products failed (first failed test): too few active months 2,099; launched too recently 686; no sale in 90 days 548; too few units 198; mostly bundled 51.

Family diagnostics

Family	Types	Products	Revenue ($M)	Margin %	Nov. index	YoY units	Eligible
Frame Road	Notebook	531	107.1	24.5%	1.36	−39%	16
Solar Peak	Notebook, Workstations	630	90.0	19.1%	1.41	−50%	5
Knight Thrive	Notebook	447	62.6	15.6%	1.16	+21%	15
Romeo Iris	Options, Services	1,383	27.7	46.4%	1.08	−16%	73
Titan Onset	Desktop	303	21.2	18.6%	0.74	−38%	3
Quest Lunar	Visuals	137	9.0	22.1%	1.23	+14%	6
Circuit Insight	Servers	73	1.9	41.9%	0.99	−93%	0
Signal Road	Options	22	0.1	−36.4%	3.29	−83%	0

Nov. index = November's share of units × 12 (1.0 = no seasonality). YoY compares Apr–Aug 2026 with Apr–Aug 2025. Signal Road sells at a loss overall. Two tiny families (Harbor Momentum, Stream Juliet) are omitted here.

Pilot 30 selection

Slots are split by product type: each type gets at least 2, and the rest follow each type's share of eligible revenue. Within a type, the top sellers by units fill its slots. This keeps the pilot laptop-heavy (laptops are 58% of eligible revenue) while still covering every category. Ranking purely by units, the PDF's rule, would give a pilot of mostly cheap accessories.

Type	Eligible	Share of eligible revenue	Pilot slots
Notebook	31	58%	17
Options	48	16%	4
Services	25	10%	3
Workstations	5	9%	2
Desktop	3	3%	2
Visuals	6	3%	2

Daily training panel

30 products × 1,246 days (37,380 rows). Columns: units, revenue, cost, lines, customers, paid ASP, last known price (the previous day's ASP), unit cost, days since last sale, day of week, month, fiscal quarter, Canadian holidays, month/quarter end, Black Friday week, coverage-gap flag, and product attributes. Gap days and days before a product's launch are blank, not zero.

Stage 1.5 — Global Baseline v1

The learned blend beats every simple benchmark overall (45.3% WAPE vs 45.9% for last-28-days). The gain comes from laptops; on accessories and services, last-28-days is still slightly better.

How the baseline works

- **Target:** units sold in the next 28 days, forecast every Monday ("origin") from Apr 2024 onward.

- **Components:** six recent sales rates, each using only days before the origin and scaled to 28 days: last 7, 28, 91, 182 and 365 days (`ma7` … `ma365`), plus the same 28 days last year (`ly28`).
- **Baseline** = weighted sum of the six components. Weights are ⩾ 0 and add up to 1.
- **Fitting:** the weights minimise total absolute error pooled across all 30 products, which is the same as minimising WAPE. This is solved exactly as a linear programme.
- **Missing history:** if a component has under 50% known days, it falls back to the next shorter window (ma365 → ma182 → ma91 → ma28; ly28 → ma91), so new laptops use the same weights.

Time-based validation

This is an expanding window: for each test quarter, the weights are learned only from forecasts whose 28-day target finished before that quarter began, then scored on the quarter. The test quarters are Jan, Apr and Jul 2025 and Jan, Apr and Jul 2026; Oct 2025 is skipped because of the coverage gap. That gives 1,462 scored product-weeks out of 1,901 usable rows.

Model	WAPE all	WAPE lifecycle	WAPE run-rate	MAE (units)	Bias
Global v1 (learned blend)	**45.3%**	**61.9%**	34.3%	44.9	+1.6%
Last 28 days	45.9%	64.8%	**33.3%**	45.5	−1.0%
Equal weights	46.2%	62.5%	35.3%	45.8	+0.4%
Last 7 days	48.6%	67.4%	36.1%	48.2	−1.2%
Last 91 days	50.5%	68.7%	38.3%	50.1	+4.5%
Last 182 days	53.3%	70.5%	41.9%	52.9	+6.4%
Last 365 days	56.0%	72.1%	45.2%	55.5	+0.7%
Same 28 days last year	65.0%	77.5%	56.6%	64.5	−7.1%

Lower is better. WAPE = total absolute error ÷ total actual units.

Learned weights

Weights learned in	Train rows	ma7	ma28	ma91	ma182	ma365	ly28
Frozen v1 (all data)	1,901	**0.34**	**0.27**	**0.27**	0.00	0.05	0.07
Jul 2026 fold	1,661	0.29	0.15	0.40	0.05	0.04	0.08

Weights learned in	Train rows	ma7	ma28	ma91	ma182	ma365	ly28
Apr 2026 fold	1,271	0.23	0.23	0.28	0.15	0.00	0.11
Jan 2026 fold	1,001	0.20	0.22	0.33	0.13	0.00	0.12
Jul 2025 fold	747	0.17	0.07	0.61	0.03	0.00	0.12
Apr 2025 fold	565	0.18	0.00	0.71	0.00	0.00	0.11
Jan 2025 fold	383	0.00	0.53	0.47	0.00	0.00	0.00

As training data grows, weight shifts from the 91-day average toward the most recent weeks, and settles at roughly one-third each on 7, 28 and 91 days. Year-long and last-year components carry little weight. The average spread of each weight across folds is 0.10.

The PDF's monthly results (last month 39.3% WAPE in Canada) aren't directly comparable: they use a monthly target and a different top 30. Both runs agree that "last month" is very hard to beat and that extra complexity buys little.

Stage 1.8 — Segmentation and Baseline v2

No segmentation improved accuracy enough to adopt, so Baseline v2 is frozen as one global profile with the v1 weights. With only 30 products, separate weights per group mostly fit noise.

What was tested

Each scheme learns its own weights per segment, using the same time-based folds as v1. A segment with fewer than 4 pilot products uses the global weights. To be adopted, a scheme had to cut WAPE by at least 2% (relative) vs v1, and keep its weights stable across folds (average spread ⩽ 0.20).

Scheme	Segments (with own weights)	WAPE	Change vs v1	Weight spread	Adopted
Lifecycle tier (lifecycle vs run-rate)	2 (2)	45.24%	−0.1%	0.09	No: gain too small
Country	1 (1)	45.30%	0.0%	0.10	Not testable: Canada only
Demand pattern (smooth / erratic / lumpy)	3 (3)	45.46%	+0.4%	0.09	No: worse
Product family	6 (3)	45.55%	+0.5%	0.11	No: worse

Scheme	Segments (with own weights)	WAPE	Change vs v1	Weight spread	Adopted
Product type	6 (2)	45.65%	+0.8%	0.16	No: worse and unstable

The demand pattern is the Syntetos–Boylan class from weekly sales (how often a product sells and how variable its order sizes are).

What the segments learned

Laptop-type segments lean on recent weeks plus the 182-day average (e.g. Knight Thrive: ma7 0.42, ma182 0.43). Accessories and services lean on the 91-day average (ma91 0.36). The lifecycle split helps laptops (61.1% vs 61.9% WAPE) but hurts run-rate products (34.7% vs 34.3%), so the net gain is near zero.

Adding a momentum component (2×ma28 − ma91, or a damped trend) was also tested; it did not change WAPE.

Next step: retest the lifecycle-tier split once the US data is added, which roughly doubles the sample. It is the leading candidate for a 2-profile Baseline v2.

Stage 2 — Final recommended price

Prices go up on thin-margin or less price-sensitive laptops, and down on higher-margin ones where volume responds strongly. The 30 products split into 10 up, 10 down and 10 holds. Expected effect over the next 28 days: about +$13.3k margin and −$5.2k revenue (as of 2026-08-28, the last data date).

Pricing logic

1. **Business signals** (per product, as of the last data date):
    - Current price: the last 28 days' ASP, or 91 days if fewer than 5 units sold.
    - Unit cost and margin %.
    - Baseline v2 units for the next 28 days.
    - Momentum (28-day vs 91-day rate) and YoY units.
    - Price position vs the product's own 365-day median.
    - Top-customer share and days since last sale.
    - **Price elasticity**:
        - A weekly log-units vs log-price regression per product, with a trend term.

- Shrunk toward the product type's pooled estimate, weighted by how precise each estimate is.
- Bounded to [−4, −0.2].
- Rated "high" confidence if its posterior SD is below 0.5.

2. **Objective weights** (placeholders, to agree with Bob): margin 0.40, revenue 0.25, volume 0.20, stability 0.15.
3. **Price adjustment:**
   - For each candidate move from −15% to +15% in 1% steps: expected units = baseline × (1 + move)^elasticity.
   - Each objective is scored as % change vs holding the price: margin $, revenue $ and units, plus stability = −|move|.
   - The move with the highest weighted score wins.
4. **Guardrails**, applied in order:
   a. No sale in 28+ days → hold and review (possible end-of-life).
   b. Fewer than 20 units in 91 days → hold.
   c. Step cap ±10% per cycle, or ±3% when elasticity confidence is low.
   d. Stay within the trailing-365-day P5–P95 price band ±5%.
   e. Price ≥ cost + 5%.
   f. Moves under 1% → hold.
   g. Round to whole dollars ($100+) or dimes, always toward the current price.

Most unconstrained answers hit the ±15% edge of the grid. That's expected with constant elasticity (revenue and volume always favour one direction), so in practice the step cap sets the move size.

Recommendations

Product	Type	Current	Recommended	Change	Elasticity	Binding guardrail	Exp. margin Δ 28d ($)
71SI047PCZ	Notebook	$1,030.47	$1,061.00	+3.0%	−2.45	step cap ±3% (low confidence)	+2,959
95SS294NWP	Notebook	$1,106.26	$1,139.00	+3.0%	−2.09	step cap ±3% (low confidence)	+2,341

Product	Type	Current	Recommended	Change	Elasticity	Binding guardrail	Exp. margin Δ 28d ($)
17D4481BEM	Notebook	$1,006.90	$1,037.00	+3.0%	−2.15	step cap ±3% (low confidence)	+2,287
89UY8827CW	Notebook	$1,040.91	$1,072.00	+3.0%	−2.79	step cap ±3% (low confidence)	+1,977
03DF114WYO	Workstations	$1,846.14	$1,901.00	+3.0%	−1.50	step cap ±3% (low confidence)	+1,960
19Y55740YG	Notebook	$878.62	$904.00	+2.9%	−2.31	step cap ±3% (low confidence)	+1,544
87RIOCI9WY	Notebook	$1,551.82	$1,598.00	+3.0%	−1.30	step cap ±3% (low confidence)	+582
41GEFQX2KE	Notebook	$1,094.49	$1,127.00	+3.0%	−1.75	step cap ±3% (low confidence)	+517
31P6JRA4LY	Notebook	$993.72	$1,023.00	+2.9%	−2.19	step cap ±3% (low confidence)	+424
06AS870MKS	Desktop	$1,010.19	$1,040.00	+3.0%	−1.26	step cap ±3% (low confidence)	+252
14XOTJR8EF	Notebook	$1,624.23	$1,576.00	−3.0%	−3.08	step cap ±3% (low confidence)	+400
4Q01B61187	Options	$19.43	$17.50	−9.9%	−2.30	step cap ±10%	+77
36Z9DAK3EW	Desktop	$1,543.66	$1,498.00	−3.0%	−3.19	step cap ±3% (low confidence)	+14
16T0MJI6WJ	Workstations	$2,030.01	$1,970.00	−3.0%	−2.48	step cap ±3% (low confidence)	−18

Product	Type	Current	Recommended	Change	Elasticity	Binding guardrail	Exp. margin Δ 28d ($)
5YO5U33136	Services	$96.40	$93.60	−2.9%	−1.86	step cap ±3% (low confidence)	−38
2B70T80159	Options	$18.05	$16.30	−9.7%	−1.96	step cap ±10%	−70
37NELHX9FG	Notebook	$1,933.36	$1,876.00	−3.0%	−3.62	step cap ±3% (low confidence)	−268
02TYQZV3PJ	Notebook	$1,472.74	$1,429.00	−3.0%	−2.15	step cap ±3% (low confidence)	−270
16CHOUW4TD	Notebook	$1,680.44	$1,631.00	−2.9%	−2.91	step cap ±3% (low confidence)	−386
02XJSFY3QN	Notebook	$1,731.57	$1,680.00	−3.0%	−2.02	step cap ±3% (low confidence)	−1,013
9TE3M17113	Services	$148.97	hold	0%	−0.69	none: objectives balance	0
21WO7047EX	Notebook	$2,693.03	hold	0%	−2.64	under 20 units in 91 days	0
27WH3132DN	Options	$144.14	hold	0%	−2.18	no sale in 88 days	0
5IQ5M98979	Services	$156.46	hold	0%	−2.01	no sale in 79 days	0
87JI6103KJ	Notebook	$929.38	hold	0%	−3.04	no sale in 74 days	0
01Y5AUP2ZB	Visuals	$226.45	hold	0%	−1.83	no sale in 74 days	0
55C97203QD	Options	$322.40	hold	0%	−2.68	no sale in 66 days	0
00RN3110HH	Notebook	$1,886.62	hold	0%	−2.51	no sale in 60 days	0

Product	Type	Current	Recommended	Change	Elasticity	Binding guardrail	Exp. margin Δ 28d ($)
13BASGW8QU	Visuals	$217.21	hold	0%	−2.20	no sale in 38 days	0
49TH570PJS	Notebook	$1,120.29	hold	0%	−2.85	no sale in 37 days	0

Increases mostly go to laptops with 10–18% margins, where a 3% price rise lifts margin dollars by more than the units it loses. Decreases go to laptops with 24–39% margins, where the extra volume is worth more under the current weights. Holds marked "no sale" are flagged as possible end-of-life for review. The "Why" text for each product is in the `rationale` column of `s2_price_recommendations.csv`.

Limitations and open decisions

Limitations

- **Elasticities are observational.** Within a week, price differences come from customer mix and volume discounts, which probably makes laptops look more price-sensitive than they are. 26 of 30 estimates are low-confidence. Confirm them with controlled price tests before acting on them.
- **Small sample.** With 30 products, segmentation couldn't beat global weights; adding the US market is the cheapest fix.
- **Survivorship bias.** The pilot was picked using the full history (as a business pilot), so its validation slightly favours products that survived.
- **Pilot churn.** 8 of the 30 pilot products haven't sold in 4+ weeks, so a fixed pilot goes stale fast.
- **Missing signals.** No competitor prices (Dell, HP, Acer) and no inventory are in the data.
- **Same customer, many prices.** B2B prices are negotiated per customer, so the "current price" is an average and not a list price.

Decisions for the team

- [ ] Agree the objective weights with Bob (currently margin 0.40, revenue 0.25, volume 0.20, stability 0.15).
- [ ] Confirm the guardrails: ±10% step, ±3% at low confidence, cost + 5% floor.

- [ ] Choose the pilot design: mixed (current) vs hardware-only, and fixed vs refreshed monthly like the PDF's dynamic top 30.
- [ ] Confirm the eligibility thresholds, especially 24 months for run-rate and 9 for lifecycle.
- [ ] Run the full pipeline on the US file and retest the lifecycle-tier profile.
- [ ] Confirm whether `revenue` is USD, and which currency recommendations should show in.

Suggested slide outline

This is a 12-slide deck. Each slide lists its title, then the message and where its content comes from in this doc.

1. **PriceWise: from transactions to a recommended price**
   - Title slide.
   - The four stages, one line each (Pipeline at a glance).
2. **Data we used and what we cleaned**
   - 498k Canada invoice lines, Apr 2023 – Aug 2026.
   - Nothing deleted, everything labelled (cleaning table).
3. **A data gap we found: Oct–Dec 2025**
   - About 2 active customers a week vs about 25 normally.
   - Treated as missing, not as zero sales.
   - Visual: median weekly active customers by source quarter.
4. **Which products can we price?**
   - 118 of 3,715 products are eligible.
   - Two lifecycle tiers, because laptops live about 12–15 months (eligibility table and fail reasons).
5. **Family diagnostics**
   - Laptops carry most of the revenue at 16–25% margins.
   - Romeo Iris (Options and Services) is high-margin.
   - Signal Road loses money.
6. **Pilot 30**
   - Revenue-weighted by product type, minimum 2 per type.
   - 17 notebooks, plus coverage of every category (pilot table).
7. **The baseline model**

- A weighted blend of six recent sales averages.
- Weights learned by minimising WAPE.
- Visual: diagram of the components feeding the weighted sum.

8. **Does it work?**
   - 45.3% WAPE vs 45.9% for last-28-days.
   - Wins on laptops (61.9% vs 64.8%).
   - Visual: bar chart of WAPE by model (benchmark table).
9. **Learned weights**
   - About one-third each on the 7-, 28- and 91-day averages.
   - Year-ago data matters little (weights table).
10. **Segmentation test**
    - No split beat global by 2%, so v2 = one profile.
    - Lifecycle tier is the one to retest with US data.
11. **From forecast to price**
    - Signals, then objective weights, then the best move, then guardrails.
    - Visual: flow diagram (Pricing logic).
12. **Recommendations and next steps**
    - 10 up, 10 down, 10 hold; +$13.3k margin and −$5.2k revenue over 28 days.
    - Decisions for Bob: objective weights, guardrails, pilot design.

Charts for slides 3, 8 and 9 can be built directly from `s1_coverage_by_quarter.csv`, `s15_validation_summary.csv` and `s15_weights_by_fold.csv`.

Files and how to rerun

Run: put `pricewise_pipeline.ipynb` next to the data, set `RAW_FILES` in the Config cell, then run all cells top to bottom (about 30 seconds). It needs `pandas numpy scipy statsmodels holidays pyarrow`; the first cell has the pip line. Every threshold in this doc lives in the Config cell. To add the US market, add its file to `RAW_FILES`; everything then runs per country.

Output file (`outputs/`)	Stage	What's in it
`s1_coverage_by_quarter.csv`	1	Active customers per source quarter; gap flag

Output file (`outputs/`)	Stage	What's in it
`s1_product_eligibility.csv`	1	Every product, each test pass/fail, first failed rule
`s1_family_diagnostics.csv`	1	One row per product family
`s1_pilot_30.csv`	1	The 30 pilot products and their attributes
`s1_daily_panel.parquet`	1	Product × day training panel
`s15_validation_summary.csv`	1.5	WAPE / MAE / RMSE / bias per model, overall and per tier
`s15_weights_by_fold.csv`	1.5	Weights learned in each test quarter
`baseline_v1.json`	1.5	Frozen Global Baseline v1
`s18_scheme_comparison.csv`	1.8	Segmentation results
`baseline_v2.json`	1.8	Frozen Baseline v2 and each product's profile
`s2_business_signals.csv`	2	All signals, including elasticity details
`s2_objective_grid.csv`	2	Score of every candidate price move
`s2_price_recommendations.csv`	2	Final prices, guardrails, expected impact, rationale